Department of Mathematics and Statistics, School of Arts and Science, Qatar University.

Unit Name:	Differential Equations	Instructor:	Dr. Mohamed Mabrok
Section:	L151	Location:	Lecture Room
Exam :	**Final Exam: Fall 2025**	Time and Date:	
Student Name:			
Student ID:			

- No credit will be given to answers for non-requested questions.
- Only non-plotting calculators are permitted. The use of the cell phones is absolutely prohibited.
- Shut off your cell phone completely.
- Make sure that you are feeling comfortable before entering the class for the test.
- Full mark will be given for complete answers for all questions.
- Cheating is an academic violation according to Qatar University rules and regulations, and in some cases, it may result in final dismissal from the University. Students should not under any circumstances commit or participate in any cheating attempt or any act that violates student code of conduct.

Question Number	Student Marks	Maximum Marks
1		**8**
2		**10**
3		**12**
4		**12**
5		**12**
6		**10**
7		**6**
Total / (Max)		**70**

1. Solve the following differential equation:

$$(y+\cos x)dx+(\sin x+xy+x)dy=0$$

8 Marks

2. Find the general solution of the following differential equation if $y_1 = e^{-x}$ satisfies the complementary equation of the following DE.

$$(2x+1)y'' - 2y' - (2x+3)y = (2x+1)^2$$

10 Marks

3. Using Laplace transforms, solve the folloing IVP

$$y'' - y' = \cos(2t - 12)\, u_6(t) \qquad y(0) = -4,\ y'(0) = 0$$

12 Marks

4. Use Laplace transforms, find a general formula for the solution of the initial value problem:

$$y'' + 2y' + 5y = f(t), \quad y(0) = 0, \quad y'(0) = 1,$$

using the convolution operation. Then, use the obtained general form and

$$f(t) = e^{-t}$$

to find the solution (*You have to compute the integration*) 12 Marks

5. Show how the equation

$$y'' + 2y' + 5y = 0$$

can be transformed to the system

$$X' = \begin{bmatrix} 0 & 1 \\ -5 & -2 \end{bmatrix} X$$

Then, solve the system using matrix method. 12 Marks

6. Find the power series in x for the general solution of

$$(1+x^2)y'' + 6xy' + 6y = 0$$

and show that it can be given as:

$$\boxed{y = a_0(1 - 3x^2 + 5x^4 - 7x^6 + \dots) + a_1(x - 2x^3 + 3x^5 - \dots)}$$

10 Marks

7. **1.** Consider the differential equation in standard form:

$$y'' + P(x)y' + Q(x)y = 0$$

If both coefficient functions $P(x)$ and $Q(x)$ are analytic at $x = 0$, how is the point $x = 0$ classified?

(a) Ordinary Point
(b) Regular Singular Point
(c) Irregular Singular Point
(d) Indicial Point

2. You calculate the roots of the indicial equation for a regular singular point and find $r_1 = \frac{3}{2}$ and $r_2 = \frac{1}{2}$. Which statement describes the form of the linearly independent solutions?

(a) This gives at least one of the two linearly independent solutions where one solution is a Frobenius series, and the second solution typically involves a logarithmic term $y_1 \ln(x)$.
(b) Both solutions are of the form $y = \sum_{n=0}^{\infty} a_n x^{n+r}$.
(c) Both solutions must be standard Taylor series of the form $\sum a_n x^n$.
(d) One solution is of the form $y_1 = x^{r_1} \sum a_n x^n$, and the second solution must strictly involve $\sqrt{x}$ as a separate multiplier.

3. If the roots of the indicial equation are identical, $r_1 = 4$ and $r_2 = 4$ (Case 3), which statement is true about the second linearly independent solution?

(a) It is simply a constant multiple of the first solution.
(b) It can be found using the exact same Frobenius form $\sum a_n x^{n+r}$ as the first solution.
(c) It will invariably involve a logarithmic term of the form $y_1(x)\ln(x) + \sum b_n x^{n+r}$.
(d) It does not exist; second-order equations with equal indicial roots have only one solution.

4. Inspect Bessel's Equation:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0$$

By converting to standard form and analyzing the limits of the coefficients as $x \to 0$, what is the classification of the point $x = 0$?

(a) Ordinary Point
(b) Regular Singular Point
(c) Irregular Singular Point
(d) Complex Singular Point

6 Marks .

Algebra

- $\sin\theta = \frac{1}{\csc\theta}$
- $\csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$
- $\sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$
- $\cot\theta = \frac{1}{\tan\theta}$
- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$
- $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$
- $\cos^2(x) = 1 - \sin^2(x)$
- $\sin^2(x) = 1 - \cos^2(x)$

Differentiation

- $\frac{\mathrm{d}}{\mathrm{d}x} c = 0$
- $\frac{\mathrm{d}}{\mathrm{d}x} x^n = n x^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} (ax \pm b)^n = a\, n\, (ax \pm b)^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|x| = \frac{1}{x}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|ax \pm b| = \frac{a}{ax \pm b}$
- $\frac{\mathrm{d}}{\mathrm{d}x} e^{ax} = a\, e^{ax}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \sin(ax) = a\, \cos(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} \cos(ax) = -a\, \sin(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\, g(x) = f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x} + g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x}$ *Product rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} \frac{f(x)}{g(x)} = \frac{g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x} - f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x}}{(g(x))^2}$ *Quotient rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)$ *Chain rule.*

Integration

- $\int a\, dx = ax + c$
- $\int x^n\, dx = \frac{x^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int f(x)^n f'(x)\, dx = \frac{f(x)^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int \frac{a}{x}\, dx = a\, \ln|x| + c$
- $\int \frac{1}{ax+b}\, dx = \frac{1}{a} \ln|ax+b| + c$
- $\int e^{ax}\, dx = \frac{1}{a}\, e^{ax} + c$
- $\int \frac{f'(x)}{f(x)}\, dx = \ln|f(x)| + c$
- $\int \sin(ax)\, dx = \frac{-1}{a}\, \cos(ax) + c$
- $\int \cos(ax)\, dx = \frac{1}{a}\, \sin(ax) + c$
- $\int \sec^2(ax)\, dx = \frac{1}{a}\, \tan(ax) + c$
- $\int \csc^2(ax)\, dx = -\frac{1}{a}\, \cot(ax) + c$
- $\int \tan(ax)\, dx = -\frac{1}{a}\, \ln(\cos(ax)) + c$
- $\int \cot(ax)\, dx = \frac{1}{a}\, \ln(\sin(ax)) + c$
- $\int \sec(ax)\tan(ax)\, dx = \frac{1}{a}\, \sec(ax) + c$
- $\int \csc(ax)\cot(ax)\, dx = -\frac{1}{a}\, \csc(ax) + c$
- $\int \sec(ax) dx = \frac{1}{a}\, \ln(\sec(ax) + \tan(ax)) + c$
- $\int \csc(ax)\, dx = \frac{-1}{a}\, \ln(\csc(ax) - \cot(ax)) + c$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$

Table of Laplace Transforms

	$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$		$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	$\mathbf{e}^{at}$	$\frac{1}{s-a}$
3.	$t^n,\quad n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p,\ p>-1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\quad n=1,2,3,\ldots$	$\frac{1\cdot 3\cdot 5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$\mathbf{e}^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$\mathbf{e}^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$\mathbf{e}^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22.	$\mathbf{e}^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n\mathbf{e}^{at},\quad n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{\mathbf{e}^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	$\mathbf{e}^{-cs}$
27.	$u_c(t)f(t-c)$	$\mathbf{e}^{-cs}F(s)$	28.	$u_c(t)g(t)$	$\mathbf{e}^{-cs}\mathfrak{L}\{g(t+c)\}$
29.	$\mathbf{e}^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\quad n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T \mathbf{e}^{-st}f(t)\,dt}{1-\mathbf{e}^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$	36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			